

Are Linear Forecasters Enough? A Small Controlled Study of DLinear, a Patch Transformer and a GRU on Synthetic Series with Trend, Seasonality, Noise and Regime Switching

Pyrite-89A

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/dlinear-vs-transformer-synthetic-forecasting>

Abstract—Linear models such as DLinear were reported to match or beat Transformer forecasters on common long-horizon benchmarks, but real datasets do not let one vary the data property that might favour nonlinear models. We compare, under one protocol, a seasonal-naive forecaster, a DLinear-style decomposition-plus-linear model, a small patch transformer (PatchTST-style) and a small patch GRU, all reimplemented on CPU, at horizons 24, 96 and 192. The data are synthetic series with controlled seasonality, trend, noise and Markov regime switching, plus ETTh1; three seeds, no hyper-parameter tuning. A win is defined in advance as at least 5% lower seed-mean MSE and lower MSE in all three seeds. The answer depends on the training budget. At the planned 400 training steps neither nonlinear model wins on any of the six tasks at any horizon; the closest case is the patch transformer on the regime-switching task (its seed-mean MSE is 0.952 times DLinear’s at horizon 96; the rule needs at most 0.95). A training-budget sweep, added after an independent audit, changes this on the regime-switching task only: at 1000 and 2000 steps both nonlinear models win at horizon 24 and the GRU wins at horizon 96 (seed-mean MSE 0.386 against 0.432 for DLinear at 2000 steps), while DLinear’s MSE barely moves with the budget; nobody wins at horizon 192. On the four tasks without regime changes (tested at horizon 96) and on ETTh1 (all horizons) no nonlinear model wins at 2000 steps either, and on ETTh1 their MSE stays above DLinear’s at horizons 96 and 192 (at 2000 steps and horizon 96: 0.403 and 0.386 against 0.375). At 2000 steps the GRU passes the rule at mean regime lengths of 100 and 500 steps but not at 250, 1000 or 4000; its margin is not monotone in the switching rate. Removing the decomposition changes DLinear’s MSE by less than the 5% registered for this check (400 steps). The evidence is small-scale: three seeds, at most 2000 steps, untuned models and synthetic generators of our own design; the noise sweep and the design ablation were run at 400 steps only.

Index Terms—long-horizon forecasting, DLinear, PatchTST, GRU, synthetic benchmarks, regime switching, reproducibility

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I. Introduction

Transformer architectures for long-horizon forecasting [1], [2] reported large gains on the ETT and related benchmarks. Zeng et al. [3] then showed that a one-layer linear model with a moving-average decomposition (DLinear) outperformed these models on most of the same benchmarks, which cast doubt on the benefit of temporal attention. Patching and channel independence [4] restored a Transformer advantage in later reports, while benchmark studies found that evaluation details matter and that simple methods are often competitive [5]. Each real dataset mixes trend, seasonalities, noise and non-stationarity in unknown proportions, so the question “are simple linear models enough?” has no single answer on real data.

This paper asks a narrower, controlled question: on series whose trend, seasonal structure, noise level and regime switching we set ourselves, which property, if any, lets a small nonlinear model beat a DLinear-style model at horizons 24, 96 and 192? We reimplement four systems (seasonal naive, DLinear-style, PatchTST-style, patch GRU), give them the same look-back, normalisation, optimiser and budget, and test five pre-registered hypotheses (Section IV) with an explicit win rule. We also run ETTh1 [1] as one real series, two sweeps (regime length and noise level), a design ablation and a training-budget sweep.

Contributions: (i) a protocol-matched comparison across six tasks and three horizons with three seeds each; (ii) a budget-dependent answer: at 400 training steps no nonlinear model passes the win rule anywhere in the main grid, whereas at 1000 and 2000 steps the nonlinear models pass it on the regime-switching task at horizons 24 and 96 (the GRU at both, the patch transformer at 24) and on no other task that we tested; (iii) sweeps showing that this margin is absent at horizon 192 and for long regimes, and that the low-budget patch-transformer margin depends

on noise; (iv) code, per-run logs and the run registry. The budget sweep was not in the original plan; it was added after an independent audit showed that the 400-step verdicts did not hold at 2000 steps. All claims are limited to the small scale described in Section VII.

II. Related Work

Transformers for forecasting. Informer [1] introduced sparse attention for long sequences and the ETT datasets; Autoformer [2] added a series-decomposition block and auto-correlation. TimesNet [6] models temporal variation in a 2D representation across several analysis tasks. iTransformer [7] applies attention across variables instead of time steps. PatchTST [4] splits each channel into patches used as tokens and treats channels independently; our patch transformer follows this design in a reduced form.

Linear and MLP models. Zeng et al. [3] proposed DLinear and NLinear, a moving-average decomposition into trend and remainder, each mapped to the horizon by one linear layer, and argued that earlier Transformer gains came from direct multi-step prediction. N-BEATS [8] is an earlier fully connected stack with basis expansion. Kim et al. [9] question the usefulness of self-attention in forecasting Transformers.

Recurrent models. The GRU [10] is a standard gated recurrent unit. xLSTMTIME [11] adapts an xLSTM recurrent architecture to long-term forecasting; our GRU is a much smaller patch-based variant.

Benchmarking. TFB [5] documents evaluation pitfalls in forecasting comparisons (for example dropped last batches) and compares many methods on a broad set of datasets; the Monash archive [12] collects forecasting datasets for evaluating methods across domains. Accordingly we use all test windows (no dropped batch) and identical settings for every system.

Gap. These works compare architectures on real datasets whose properties are not controlled. We propose no new model; we vary one data property at a time with a fixed pipeline, at a scale a single CPU can reproduce.

III. Method

All systems forecast one channel at a time (channel independent). Given a look-back window $x \in \mathbb{R}^L$ ($L = 336$) a model predicts the next $H \in \{24, 96, 192\}$ values; a separate model is trained per horizon.

Instance normalisation. For the learned models, $\mu = \text{mean}(x)$, $\sigma = \text{std}(x) + 10^{-5}$ over the window, $\tilde{x} = (x - \mu)/\sigma$, and the output is $\hat{y} = f(\tilde{x})\sigma + \mu$. It is applied identically to all three learned models and switched off in one ablation.

Seasonal naive (reimplemented). $\hat{y}_{t+h} = x_{t-m+1+(h-1) \bmod m}$, $h = 1..H$, i.e. the last full cycle of length m is repeated; $m \in \{24, 168\}$ is chosen per task, horizon and seed on the validation split.

DLinear-style (reimplemented). With a moving average of kernel $k = 25$ and replicate padding, $T = \text{MA}_k(\tilde{x})$, $R = \tilde{x} - T$, and

$$\hat{y} = (W_R R + b_R + W_T T + b_T)\sigma + \mu, \quad W_R, W_T \in \mathbb{R}^{H \times L}, \quad (1)$$

with weights shared across channels and initialised to $1/L$. The “no decomposition” variant uses $\hat{y} = (W\tilde{x} + b)\sigma + \mu$.

PatchTST-style (reimplemented). Patches of length 16 with stride 8 give $N = 41$ tokens $p_i \in \mathbb{R}^{16}$, embedded as $z_i^0 = W_e p_i + e_i$ with learned positions e_i ; two pre-norm encoder layers ($d = 64$, 4 heads, feed-forward width 128, dropout 0.1) compute $z^{l+1} = z^l + \text{MSA}(\text{LN}(z^l))$ followed by $z^{l+1} \leftarrow z^{l+1} + \text{FF}(\text{LN}(z^{l+1}))$. The head is linear on the flattened tokens: $\hat{y} = (W_o \text{vec}(z^2) + b_o)\sigma + \mu$.

Patch GRU. Non-overlapping patches of 8 values form a sequence of 42 inputs to a one-layer GRU with hidden size 64, $h_j = \text{GRU}(h_{j-1}, p_j)$, and $\hat{y} = (W_o h_{42} + b_o)\sigma + \mu$.

Training. Mean squared error, AdamW (weight decay 10^{-4}), cosine learning-rate decay over the whole budget, batch 64 windows sampled uniformly with random channel, gradient clipping 1, learning rate $3 \cdot 10^{-3}$ (DLinear, GRU) and 10^{-3} (PatchTST), fixed in advance. The budget is 400 steps in the main grid, the design ablation and the dwell and noise sweeps; the budget sweep repeats runs with 1000 and 2000 steps and no other change. Validation MSE (every eighth validation window) is computed every 100 steps and the best checkpoint is kept, so a longer budget also means more candidate checkpoints, for every system alike.

Win rule. For a nonlinear system s and DLinear on a task and horizon, with seed-mean MSE $\bar{e}_s, \bar{e}_D$ over the same seeds,

$$\text{win}(s) \iff (\bar{e}_s - \bar{e}_D)/\bar{e}_D \leq -5\% \quad \text{and} \quad e_s^{(i)} < e_D^{(i)} \quad \forall i. \quad (2)$$

Equivalently, the ratio $\bar{e}_s/\bar{e}_D$, which is what we report for the main grid, must be at most 0.95. This rule was fixed in the proposal before running and is applied unchanged at every budget; with three seeds it is a screening rule, not a significance test.

IV. Experimental Setup

Tasks. Each synthetic dataset has 4 channels of 8000 steps (60/20/20 chronological train/validation/test); its generator seed is 1000+seed, so a seed is a new dataset (shared by all systems) as well as a new initialisation. A channel is $y_t = \sum_k A_k s_t \sin \phi_{k,t} + \text{trend}_t + e_t$ with integrated phase $\phi_{k,t} = \phi_{k,0} + \sum_{u \leq t} 2\pi/P_{k,u}$ and AR(1) noise e_t ($\rho = 0.5$, stationary sd = “noise”). seas1: one period 24, noise 0.3. trend: seas1 plus a piecewise-linear trend (slope redrawn every 600–1400 steps). multiseas: periods 24/168/12 with amplitudes 1/0.7/0.4. noisy: multiseas with noise 1.5. regime: multiseas where a Markov chain over three regimes (geometric dwell, mean 500 steps) rescales periods by 1/1.6/0.6 and amplitudes. etth1: the public ETTh1 CSV, first 14400 rows, 7 channels,

split 8640/2880/2880; seeds change only initialisation and batch sampling. All data are z-scored per channel with training statistics.

Metrics. Test MSE and MAE in z-scored units over all windows with stride 1 (no dropped batch), pooled over channels. Mean $\pm$ sd over 3 seeds (0, 1, 2).

Experiments. Main grid: 4 systems $\times$ 6 tasks $\times$ 3 seeds (72 runs, each covering the 3 horizons with separately trained models). Design ablation: no instance norm for each learned model and no decomposition for DLinear, $H = 96$, tasks trend/regime/etth1 (36 runs). Sweeps on the regime task at $H = 96$: mean dwell $\in \{100, 250, 1000, 4000\}$ (36 runs) and noise sd $\in \{0.1, 0.6, 1.5, 3.0\}$ (36 runs); the default dwell 500 and noise 0.3 rows are the main-grid regime rows at $H = 96$ (models are trained per horizon with the same seed, so these equal a separate $H = 96$ run). No hyper-parameter was tuned on any split.

Training-budget sweep (post hoc). After an independent audit found that the 400-step verdicts changed at 2000 steps, we added runs with the same code and settings except the number of steps: (a) regime and etth1 at 1000 and 2000 steps, all three horizons; (b) seas1, trend, multiseas and noisy at 2000 steps, $H = 96$ only (72 runs for (a) and (b)); (c) the dwell sweep repeated at 2000 steps, $H = 96$ (36 runs). Each cell has the three learned systems and three seeds; the 400-step cells are the main-grid rows. The noise sweep and the design ablation were not repeated. This sweep is not in the registered plan; the win rule applied to it is.

Hardware and runtime. A shared CPU machine, 2 threads per process, at most two processes at a time. Parameter counts and the median per-run wall-clock on the regime task (under load from other jobs, so only indicative; wall-clock is the one logged metric that does not repeat from run to run) are in Table I; two main-grid PatchTST runs exceeded six minutes under heavy machine load (durations are in the registry), all others stayed below. Code, runners, per-run logs and the registry (results/runs.jsonl) are in the repository. Two discard-and-rerun events preceded the reported runs (both recorded with rh decide): the first launch labelled every run with one system name (a shell bug in the runner), and stray processes from that launch then wrote further mislabelled rows. Both times the affected registry rows and raw files were removed and all runs redone; logs of the discarded runs were not kept, so the registry lists the runs that were kept, not every process started.

V. Results

Tables II and III give the planned 400-step grid, where no cell passes the win rule; Table IV adds 1000 and 2000 steps, where some regime cells pass it. Figure 1 summarises $H = 96$.

Effect of the budget. DLinear is converged at 400 steps: on regime and etth1 its seed-mean MSE is almost the same

TABLE I

Model size and median wall-clock per run over the three seeds of the regime task (3 horizons, loaded machine) at 400 steps (main grid) and 2000 steps (budget sweep).

System	Params ($H=96$)	Median run (s)	
		400 steps	2000 steps
Seasonal naive	—	0.2	—
DLinear	64704	2.2	7.0
PatchTST	322656	54.1	205.3
GRU	20448	14.1	49.1

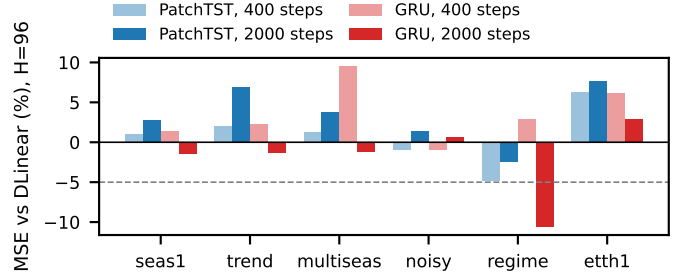


Fig. 1. MSE at $H = 96$ relative to DLinear per task (3-seed means) at 400 (light) and 2000 (dark) training steps; dashed line: the 5% win threshold.

at 400, 1000 and 2000 steps (Table IV; e.g. regime $H = 96$: 0.434, 0.434 and 0.432). The GRU is not: on regime its MSE falls from 0.322 to 0.253 at $H = 24$ and from 0.446 to 0.386 at $H = 96$. The patch transformer improves at regime $H = 24$ (0.291 to 0.277) but is slightly worse at 2000 than at 400 steps in most other cells (e.g. etth1 $H = 192$, 0.438 to 0.447), which may be over-fitting (untested). The 400-step grid thus compared a converged linear model with an under-trained GRU; verdicts are given per budget.

H1 (no regime change $\Rightarrow$ DLinear not beaten): supported where tested. At 400 steps, on seas1, trend, multiseas and noisy, no nonlinear model comes near a win at any horizon: the lowest MSE ratio to DLinear in these rows of Table III is 0.9889 (GRU on noisy, $H = 24$) and the highest is 1.1111 (GRU on multiseas, $H = 24$). The small differences are not seed noise: a seed is a dataset shared by all systems, and the paired differences have the same sign in all three seeds in every one of these cells (Table III: both nonlinear models worse on seas1, trend and multiseas, both better on noisy). They are consistent but far from the 5% threshold. At 2000 steps ($H = 96$, Table IV; ratios to DLinear from rh compare on the 2000-step runs of these four tasks) there is again no win, but the signs change: the GRU's multiseas ratio goes from 1.096 to 0.988 (better in 3/3 seeds), it is 0.985 on seas1 (3/3), and both nonlinear models lose their small edge on noisy (1.014 and 1.006); PatchTST is worse than DLinear in all seeds on all four tasks (ratios from 1.014 to 1.069). H1 was not tested at 2000 steps for $H = 24$ and 192.

H2 (regime switching $\Rightarrow$ a nonlinear model wins at $H = 96$ and 192, and the margin grows with switching

TABLE II

Main results at 400 training steps: test MSE (z-scored units), mean $\pm$ sd over 3 seeds. Underline marks the second-lowest mean in a task (the lowest is set in bold, which this font may not render); the marking is by mean only and many of these differences are well inside one sd (for the synthetic tasks the sd is mostly between-dataset spread, since each seed is a new dataset).

Method	Task	mse_24 $\downarrow$	mse_96 $\downarrow$	mse_192 $\downarrow$	n
Seasonal naive	regime	1.394 $\pm$ 0.494	1.587 $\pm$ 0.320	1.584 $\pm$ 0.417	3
GRU	regime	0.322 $\pm$ 0.046	0.446 $\pm$ 0.052	0.549 $\pm$ 0.071	3
PatchTST	regime	0.291 $\pm$ 0.014	0.413 $\pm$ 0.030	0.546 $\pm$ 0.059	3
DLinear	regime	<u>0.306 $\pm$ 0.015</u>	<u>0.434 $\pm$ 0.031</u>	0.538 $\pm$ 0.045	3
Seasonal naive	etth1	0.424 $\pm$ 0.000	0.512 $\pm$ 0.000	0.581 $\pm$ 0.000	3
GRU	etth1	0.331 $\pm$ 0.002	<u>0.399 $\pm$ 0.004</u>	<u>0.438 $\pm$ 0.005</u>	3
PatchTST	etth1	0.320 $\pm$ 0.003	0.400 $\pm$ 0.009	0.438 $\pm$ 0.005	3
DLinear	etth1	<u>0.324 $\pm$ 0.001</u>	0.376 $\pm$ 0.001	0.410 $\pm$ 0.000	3
Seasonal naive	trend	0.051 $\pm$ 0.050	0.056 $\pm$ 0.056	0.071 $\pm$ 0.073	3
GRU	trend	0.029 $\pm$ 0.028	0.032 $\pm$ 0.031	0.040 $\pm$ 0.037	3
PatchTST	trend	0.028 $\pm$ 0.028	0.032 $\pm$ 0.031	0.039 $\pm$ 0.038	3
DLinear	trend	0.028 $\pm$ 0.027	0.031 $\pm$ 0.030	0.038 $\pm$ 0.037	3
Seasonal naive	multiseas	0.198 $\pm$ 0.006	0.198 $\pm$ 0.006	0.198 $\pm$ 0.006	3
GRU	multiseas	0.114 $\pm$ 0.002	0.114 $\pm$ 0.005	0.116 $\pm$ 0.003	3
PatchTST	multiseas	0.105 $\pm$ 0.002	0.106 $\pm$ 0.002	0.106 $\pm$ 0.002	3
DLinear	multiseas	0.103 $\pm$ 0.002	0.104 $\pm$ 0.002	0.104 $\pm$ 0.002	3
Seasonal naive	noisy	1.481 $\pm$ 0.047	1.485 $\pm$ 0.049	1.486 $\pm$ 0.049	3
GRU	noisy	0.773 $\pm$ 0.014	0.785 $\pm$ 0.017	0.784 $\pm$ 0.017	3
PatchTST	noisy	0.779 $\pm$ 0.014	0.785 $\pm$ 0.019	0.787 $\pm$ 0.017	3
DLinear	noisy	0.782 $\pm$ 0.014	0.793 $\pm$ 0.018	0.792 $\pm$ 0.017	3
Seasonal naive	seas1	0.304 $\pm$ 0.006	0.309 $\pm$ 0.011	0.309 $\pm$ 0.011	3
GRU	seas1	0.160 $\pm$ 0.004	0.160 $\pm$ 0.003	0.160 $\pm$ 0.002	3
PatchTST	seas1	0.159 $\pm$ 0.003	0.160 $\pm$ 0.003	0.159 $\pm$ 0.003	3
DLinear	seas1	0.156 $\pm$ 0.003	0.158 $\pm$ 0.004	0.158 $\pm$ 0.003	3

TABLE III

400 steps: seed-mean MSE of PatchTST-style and GRU divided by DLinear's (below 1 = better than DLinear) per horizon, as computed by rh compare. Superscript: number of seeds (of 3) in which the model is better than DLinear. A * would mark a win under the pre-registered rule (ratio at most 0.95 and better in all seeds); none occurs.

Task	PatchTST vs DLinear			GRU vs DLinear		
	24	96	192	24	96	192
seas1	1.0168 ⁰	1.0094 ⁰	1.0091 ⁰	1.0263 ⁰	1.0141 ⁰	1.0134 ⁰
trend	1.0238 ⁰	1.0202 ⁰	1.0221 ⁰	1.0371 ⁰	1.0228 ⁰	1.0384 ⁰
multiseas	1.0213 ⁰	1.0131 ⁰	1.0123 ⁰	1.1111 ⁰	1.0956 ⁰	1.1087 ⁰
noisy	0.9961 ³	0.9908 ³	0.9932 ³	0.9889 ³	0.9902 ³	0.9894 ³
regime	0.9503 ³	0.9518 ³	1.0141 ²	1.0503 ¹	1.0291 ¹	1.0200 ¹
etth1	0.9878 ³	1.0632 ⁰	1.0682 ⁰	1.0228 ⁰	1.0616 ⁰	1.0679 ⁰

rate): not supported as registered at any budget; partly supported from 1000 steps. At 400 steps PatchTST is better than DLinear in all three seeds at $H = 24$ (MSE ratio to DLinear 0.9503) and $H = 96$ (0.9518), both short of the 0.95 that the 5% rule requires, and is worse at $H = 192$ (1.0141); the GRU is worse on the mean at every horizon. At 1000 and 2000 steps the picture changes (Table IV): at $H = 24$ both models win (PatchTST 0.276 and 0.277, GRU 0.266 and 0.253, against 0.306 and 0.305 for DLinear) and at $H = 96$ the GRU wins (0.402 and 0.386 against 0.434 and 0.432), while PatchTST at $H = 96$ does not reach the threshold (0.418 and 0.421, better in 3/3 seeds). At $H = 192$ nobody wins at any budget: the GRU is slightly lower on the mean but better in only 2/3 seeds (0.527 and 0.530 against 0.539 and 0.538) and PatchTST is worse than DLinear. The first clause is thus

met at $H = 96$ but not at $H = 192$. The second clause is tested by the dwell sweeps (Table VI). At 400 steps the PatchTST margin is not monotone in the dwell (PatchTST against DLinear: 0.773 vs 0.774 at dwell 100, 0.258 vs 0.272 at 1000, 0.307 vs 0.312 at 4000); the GRU margin does move in the predicted direction (from 0.743, better than DLinear in 3/3 seeds, at dwell 100 to 0.599, far worse, at 4000) but never reaches a win. At 2000 steps the GRU wins at dwell 100 (0.726 vs 0.770) and 500 (0.386 vs 0.432), is lower at 250 (0.503 vs 0.552) but better in only 2/3 seeds, and has no win at 1000 (0.260 vs 0.270) or 4000 (0.347 vs 0.310): its wins are confined to mean regime lengths up to 500 steps, but the margin peaks at 500 and does not grow monotonically with the switching rate. PatchTST has no win in the 2000-step dwell sweep.

H3 (more noise erases the nonlinear advantage): mixed; tested at 400 steps only. Table VII: PatchTST wins under the rule at noise sd 0.1 (0.304 against 0.335 for DLinear) and at no higher noise level (0.413 vs 0.434 at 0.3, 0.611 vs 0.619 at 0.6, 0.922 vs 0.929 at 1.5 and 1.012 vs 1.023 at 3.0), so its above-threshold advantage disappears, but a small advantage in 3/3 seeds remains at every noise level. The GRU runs against the hypothesis: it is worse than DLinear at 0.1 (0.363) and 0.3 (0.446), close to it at 0.6 (0.616) and 1.5 (0.928), and better in 3/3 seeds at 3.0 (1.002). A shrinking relative gap is expected in any case, since all MSEs approach the noise variance.

H4 (ETTh1: DLinear not beaten at $H = 96, 192$; all learned models beat seasonal naive): supported at all three budgets. At 400 steps DLinear has MSE 0.376 and 0.410

TABLE IV

Training-budget sweep (post hoc): seed-mean test MSE; superscript = number of seeds (of 3) in which the model is better than DLinear; * = win under the registered rule. The 400-step rows are the main grid (their ratios to DLinear are in Table III).

Task	H	steps	DLinear	PatchTST	GRU
regime	24	400	0.306	0.291 ³	0.322 ¹
regime	24	1000	0.306	0.276 ^{3*}	0.266 ^{3*}
regime	24	2000	0.305	0.277 ^{3*}	0.253 ^{3*}
regime	96	400	0.434	0.413 ³	0.446 ¹
regime	96	1000	0.434	0.418 ³	0.402 ^{3*}
regime	96	2000	0.432	0.421 ³	0.386 ^{3*}
regime	192	400	0.538	0.546 ²	0.549 ¹
regime	192	1000	0.539	0.547 ¹	0.527 ²
regime	192	2000	0.538	0.550 ¹	0.530 ²
etth1	24	400	0.324	0.320 ³	0.331 ⁰
etth1	24	1000	0.320	0.313 ³	0.317 ³
etth1	24	2000	0.323	0.315 ³	0.311 ³
etth1	96	400	0.376	0.400 ⁰	0.399 ⁰
etth1	96	1000	0.376	0.399 ⁰	0.385 ⁰
etth1	96	2000	0.375	0.403 ⁰	0.386 ⁰
etth1	192	400	0.410	0.438 ⁰	0.438 ⁰
etth1	192	1000	0.410	0.441 ⁰	0.423 ⁰
etth1	192	2000	0.409	0.447 ⁰	0.437 ⁰
seas1	96	400	0.158	0.160 ⁰	0.160 ⁰
seas1	96	2000	0.159	0.163 ⁰	0.156 ³
trend	96	400	0.031	0.032 ⁰	0.032 ⁰
trend	96	2000	0.031	0.033 ⁰	0.031 ²
multiseas	96	400	0.104	0.106 ⁰	0.114 ⁰
multiseas	96	2000	0.105	0.109 ⁰	0.104 ³
noisy	96	400	0.793	0.785 ³	0.785 ³
noisy	96	2000	0.784	0.795 ⁰	0.789 ¹

at $H = 96$ and 192; the MSE ratios to DLinear are 1.0632 and 1.0682 for PatchTST and 1.0616 and 1.0679 for the GRU (Tables II, III). At 2000 steps both are again worse in all seeds (PatchTST 0.403 and 0.447, GRU 0.386 and 0.437, against 0.375 and 0.409 for DLinear; Table IV). At $H = 24$ both are slightly better in all seeds at 2000 steps (0.315 and 0.311 against 0.323), below the threshold. Every learned model beats seasonal naive (0.424, 0.512, 0.581) at all horizons and budgets.

Where the nonlinear models fail. (i) Short budget: at 400 steps the GRU is the worst learned model on multiseas at every horizon and on etth1 at $H = 24$ (at $H = 96$ and 192 it is tied with PatchTST, 0.399 vs 0.400 and 0.438 vs 0.438); the multiseas gap vanishes at 2000 steps, so it reflects under-training. (ii) Long horizon: no nonlinear model wins at $H = 192$ on regime data at any budget. (iii) Long regimes: at dwell 4000 the GRU is worse than DLinear at both budgets (0.347 vs 0.310 at 2000 steps, 0.599 vs 0.312 at 400). (iv) ETTh1: both nonlinear models are worse at $H = 96$ and 192 at every budget. Seasonal naive is far worse than every learned model on regime (1.587 vs 0.434 for DLinear at $H = 96$).

VI. Ablations and Analysis

Design ablation (H5 and normalisation; 400 steps). Table V removes the decomposition (DLinear only) or the

TABLE V

Design ablation at $H = 96$, 400 steps (test MSE). “Full” is the main-grid model and “No norm” the same model without instance normalisation, each as the mean and as the median over the three seeds; the last column is DLinear without decomposition (seed mean).

Task	System	seed mean		seed median		No decomp (seed mean)
		Full	No norm	Full	No norm	
trend	DLinear	0.031	0.052	0.0168	0.0176	0.031
trend	PatchTST	0.032	0.101	0.0170	0.0253	–
trend	GRU	0.032	5.646	0.0173	0.1006	–
regime	DLinear	0.434	0.410	0.4348	0.4069	0.435
regime	PatchTST	0.413	0.405	0.4175	0.4017	–
regime	GRU	0.446	0.430	0.4634	0.4387	–
etth1	DLinear	0.376	0.382	0.3757	0.3832	0.383
etth1	PatchTST	0.400	0.399	0.4049	0.3945	–
etth1	GRU	0.399	0.440	0.3970	0.4344	–

instance normalisation (each learned model) at $H = 96$; “Full” is the main-grid row. H5 is supported: a single linear map without decomposition leaves DLinear’s MSE almost unchanged: 0.031 against 0.031 (trend), 0.435 against 0.434 (regime) and 0.383 against 0.376 (etth1), each change within 5%. Our trends are mild and the window normalisation already removes the level, so this does not show the decomposition is useless in general.

Instance normalisation matters more, and not in one direction. On the trend task removing it raises the seed-mean MSE of DLinear from 0.031 to 0.052, of PatchTST from 0.032 to 0.101 and of the GRU from 0.032 to 5.646, but all three means are driven by one dataset (seed 1, whose trend MSE is several times that of the other two seeds). The median over seeds moves from 0.0168 to 0.0176 for DLinear, from 0.0170 to 0.0253 for PatchTST and from 0.0173 to 0.1006 for the GRU: the two nonlinear models degrade strongly on a typical seed, DLinear only mildly. On regime, in contrast, removing the normalisation lowers the seed-mean MSE for all three models (0.434 to 0.410 DLinear, 0.413 to 0.405 PatchTST, 0.446 to 0.430 GRU; the seed medians in Table V also fall), and on etth1 it hurts the GRU (0.399 to 0.440) and changes the other two little (0.376 to 0.382 and 0.400 to 0.399). The regime result is exploratory (not a registered hypothesis); window-statistics normalisation may hurt when the signal scale changes between regimes, but we did not test that.

Regime-length and noise sweeps. Tables VI and VII and Fig. 2 (regime task, $H = 96$, 3 seeds per cell) are discussed under H2 and H3. Three cautions apply. At dwell 4000 a series of 8000 steps contains at most a few regime changes, so that cell depends on which regimes fall in the test split, and its spread across seeds is large (Fig. 2). The single 400-step sweep win for PatchTST at dwell 1000 (0.258 against 0.272 for DLinear) is not robust to the budget: at 2000 steps the same cell is 0.266 against 0.270, better in only 2/3 seeds. The dwell and noise axes were each swept with the other at its default, so interactions are unknown.

TABLE VI

Dwell sweep (regime task, $H = 96$) at 400 steps (top) and 2000 steps (bottom): seed-mean MSE; superscript = number of seeds (of 3) in which the nonlinear model is better than DLinear; * = win. Dwell 500 is the main-grid row (top) or the budget-sweep row (bottom).

dwell	DLinear	PatchTST	GRU
100	0.774	0.773 ¹	0.743 ³
250	0.554	0.529 ²	0.537 ²
500	0.434	0.413 ³	0.446 ¹
1000	0.272	0.258 ^{3*}	0.337 ⁰
4000	0.312	0.307 ³	0.599 ⁰

dwell	DLinear	PatchTST	GRU
100	0.770	0.785 ⁰	0.726 ^{3*}
250	0.552	0.548 ²	0.503 ²
500	0.432	0.421 ³	0.386 ^{3*}
1000	0.270	0.266 ²	0.260 ²
4000	0.310	0.311 ¹	0.347 ¹

TABLE VII

Noise sweep (regime task, $H = 96$, 400 steps), same layout. Noise 0.3 is the main-grid row.

noise	DLinear	PatchTST	GRU
0.1	0.335	0.304 ^{3*}	0.363 ¹
0.3	0.434	0.413 ³	0.446 ¹
0.6	0.619	0.611 ³	0.616 ¹
1.5	0.929	0.922 ³	0.928 ²
3	1.023	1.012 ³	1.002 ³

VII. Limitations

Budget and tuning. The registered 400-step budget was too short for the GRU; the verdicts changed at 1000 and 2000 steps and may change again beyond 2000. The budget sweep is post hoc; it covers all horizons only for regime and etth1 and $H = 96$ for the other four tasks, and the noise sweep and design ablation exist at 400 steps only. Learning rates and model sizes were fixed in advance and never tuned, so a tuned model of either family could change the ranking. Seeds. Three seeds per cell make the win rule a coarse screen: sign-consistent effects below 5% (PatchTST on regime at 400 steps) are reported as sub-threshold, and there are no formal significance tests. Synthetic design. All synthetic conclusions depend on our own generators (sinusoids, a three-state regime process that rescales periods and amplitudes, mild trends, AR(1) noise). Other nonlinearities (state-dependent dynamics, cross-channel interactions, heavy tails) are absent, and channels and models are independent per channel. Reimplemented baselines. The patch transformer and GRU are small reimplementations, not the published configurations; a result here says nothing about the original PatchTST. One real dataset. ETTh1 is a single series; seeds vary only initialisation and batch order. Confounds. Instance normalisation is used for all learned models and changes MSE on regime and trend by amounts comparable to the model gaps (Table V), so the ranking could depend on

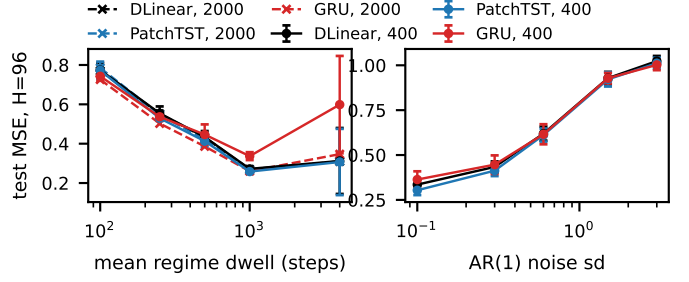


Fig. 2. Test MSE at $H = 96$ on the regime task versus mean regime dwell (left) and noise sd (right, 400 steps only). Solid: 400 steps, mean $\pm$ sd over 3 seeds; dashed: 2000 steps, mean.

it. Status of the sweeps. The dwell and noise sweeps were planned in the brief as the tests of H2 and H3, but the win checks on individual sweep cells are not registered hypotheses. A full study would add seeds, tuned baselines, budgets to convergence in every cell, more real datasets and crossed sweeps.

VIII. Conclusion

In this small CPU study the answer depended on the training budget and on one data property. At the planned 400 steps, no nonlinear model passed the pre-registered rule against a DLinear-style model on any of six tasks at horizons 24, 96 or 192. At 1000 and 2000 steps, with DLinear’s MSE essentially unchanged, the patch GRU passed it on regime-switching data at horizons 24 and 96 (seed-mean MSE 0.386 against 0.432 at $H = 96$, 2000 steps) and the patch transformer at horizon 24; neither passed it at horizon 192. Regime switching was the only property we varied under which a nonlinear model passed the rule: on the four tasks without regime changes (tested at $H = 96$ with 2000 steps) and on ETTh1 none did, and on ETTh1 DLinear remained better at $H = 96$ and 192 at every budget. The GRU’s wins were confined to mean regime lengths up to 500 steps and its margin did not grow monotonically with the switching rate, so H2 is only partly supported. Whether this holds with tuned or larger models, more seeds or longer budgets remains open.

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